Hazem Ilahi

Data Science & AI Engineering Student — Machine Learning | MLOps | LLM / RAG

Tunis, Tunisia | +216 97 111 410 | hazem.ilahi@esprit.tn | linkedin.com/in/hazem-ilahi-016a7a315 | github.com/Dyaus2

Seeking 6-month end-of-studies internship (PFE), from Jan/Feb 2027 — open to relocation

Mandatory curricular internship — tripartite internship agreement signed by ESPRIT

PROFILE

Final-year engineering student in Data Science & AI at ESPRIT (graduating 2027) with hands-on experience in machine learning, MLOps and LLM/RAG systems. During my last internship I designed the ML detection layer of a digital-forensics framework: anomaly detection plus a calibrated gradient-boosting classifier that raised attack-stage coverage from 18% to 84% at 1% false positives, shipped with a weekly retraining pipeline (champion/challenger, drift monitoring, automatic rollback). Looking for an applied data science / ML engineering internship: anomaly detection, predictive analytics, LLM/RAG applications, MLOps, forecasting.

EDUCATION

ESPRIT — École Supérieure Privée d'Ingénierie et de Technologies 2023 – 2027 (expected)

Engineering degree program, Data Science & Artificial Intelligence — final year; 6-month mandatory end-of-studies internship (PFE) from Jan/Feb 2027 *Ariana, Tunisia*

IPEIT — Preparatory Institute for Engineering Studies of Tunis 2020 – 2023

Two-year preparatory program for national engineering-school entrance exams, Physics–Chemistry *Montfleury, Tunis, Tunisia*

EXPERIENCE

Machine Learning Intern — NEXT STEP IT (Security Operations Center) Jul 2026 – Sep 2026

ATOR DFIR — ML threat-detection layer and MLOps pipeline *Charguia 1, Tunisia*

- Raised attack-stage coverage on a public adversary-emulation corpus from 18% (Sigma rules alone) to 84% (rules + ML) at a 1% false-positive rate, recovering 135 of the 168 stages missed by rules.
- Designed and integrated the ML layer of ATOR DFIR, a forensics and threat-hunting framework for Windows/Linux endpoints, on top of the security team's YARA/Sigma rule engine.
- Threat-likelihood classifier (histogram gradient boosting, Platt calibration): PR-AUC 0.927 vs 0.278 for rules, 1.6% calibration error, 5-fold cross-validation grouped by capture.
- Unsupervised anomaly detection (Isolation Forest, 300 trees): 23 of the top 25 flags on the benchmark were true attack stages; MITRE ATT&CK tactic classifier with confidence threshold (79.7% precision).
- Built the training set from 115 public attack captures (Empire, Covenant, Mimikatz, Metasploit) via ETL into the framework schema, labeled by process lineage; 110 behavioral features per process, computed by one code path shared by training and production.
- Weekly MLOps pipeline: snapshot, ETL, retraining, 14 offline quality gates, 7-day silent champion/challenger trial on live traffic, atomic deployment with canary check, automatic rollback, drift monitoring (PSI) and protection of retraining against reference poisoning.
- Integration: 10 REST endpoints and a threat-hunting page for SOC analysts (P1–P4 priorities, verdicts); 500+ automated tests, including fault-injection tests guaranteeing that an ML failure never interrupts rule-based detection.

Stack: Python 3.13, scikit-learn, pandas, NumPy, SQLite, FastAPI, Jinja2, pytest, Git

Software Developer Intern — NEXT STEP IT Jul 2025 – Sep 2025

SOC report-generation web platform (pair internship) *Charguia 1, Tunisia*

- Co-designed and developed, in a pair, a Python/Django web platform for SOC analysts with an intuitive UI/UX.
- Automated SOC report generation, replacing repetitive manual writing; prepared the solution for future deployment.

Stack: Python, Django, HTML/CSS, Git

PROJECTS

Garage Augmenté — Multi-agent RAG assistant for automotive pre-diagnosis 2026 – 2027

ESPRIT team project — owner of the knowledge base and retrieval layer

- Modular RAG pipeline on Qdrant: multilingual query expansion (French, English, Tunisian Arabic), hybrid dense BGE-M3 + BM25 retrieval fused by reciprocal rank fusion, brand/model/year filtering, cross-encoder reranking and exact lookup of 9,533 OBD fault codes.
- 7 resumable Python collectors (XML catalogs, JSON APIs, sitemaps, Next.js data routes): 2,019 unique owner's manuals (≈697,000 pages, 26 brands), 73% from official manufacturer sources, covering 54 of 56 priority models chosen from a Tunisian car-market study (2015–2026 registrations).
- Corpus preparation: SHA-256 deduplication, Tesseract OCR of ≈18,500 pages, vehicle metadata normalized into 364 model families, topic tagging; main index of 98,000 units (≈39M tokens).
- FastAPI service for the n8n workflow, ablation benchmark (hit@k, MRR) and documented hand-off package (Docker Compose, ingestion and evaluation scripts) for the GPU host.

Stack: Python, Qdrant, BGE-M3, BM25, PyMuPDF, Tesseract, FastAPI, Docker Compose, n8n

Kadastra — AI real-estate intelligence platform 2026

ESPRIT team project (Data Crunchers) — Microsoft TDSP methodology

- End-to-end pipeline for the Tunisian real-estate market: collection of ≈26,000 listings (Mubawab, Tayara), XGBoost price estimation, contract and investment-scenario generation.

- Removed a data leak in the model features; calibrated financial constants on Tunisian legislation and Central Bank (BCT) rates; full-stack deployment with FastAPI, Django, React and Docker.

Show, Attend and Tell — Image captioning (computer vision + NLP) 2026

Deep learning project — PyTorch, Flickr8k

- Reproduced the CNN + LSTM attention encoder-decoder, then extended it with a ViT encoder, CLIP reranking of beam search and BLIP fine-tuning; evaluated with BLEU and attention-map visualization.

DressMe — Smart personal fashion assistant for the Tunisian market 2026 – 2027

ESPRIT team project (Advanced Data Science Project) — design phase, Design Thinking

- User research (17-respondent survey, 6 personas) turned into 5 business and data-science objectives with metrics and a 3-version roadmap; designed a 5-layer architecture (FastAPI gateway, Redis Streams, PostgreSQL/FAISS).
- Specified 6 ML pipelines (YOLO + SAM 2 + Qwen3-VL ingestion, BPR outfit compatibility on Polyvore, Two-Tower + LightGBM recommender, ReAct resale agent) and an explainability strategy (TreeSHAP, LIME, counterfactuals).

Time-series forecasting — U.S. construction spending 2026

Statistical modeling project in R

- Three-phase analysis of residential construction spending (U.S. Census Bureau): graphical analysis, deterministic polynomial trends, ARIMA/SARIMA modeling with diagnostics and forecasts.

TECHNICAL SKILLS

Programming: Python, R, SQL, C/C++, Java, PHP
ML / DL: scikit-learn, XGBoost, PyTorch, TensorFlow, Keras; anomaly detection, classification, probability calibration, grouped cross-validation; computer vision (CNN, ViT, CLIP, BLIP); time-series forecasting (ARIMA/SARIMA)
LLM / RAG / NLP: RAG pipelines, Qdrant, BGE-M3 embeddings, BM25, hybrid search, cross-encoder reranking, retrieval evaluation (hit@k, MRR), Transformers
MLOps: automated retraining, model registry, champion/challenger deployment, drift monitoring (PSI), quality gates, canary deployment and rollback, pytest
Data: pandas, NumPy, ETL, web scraping, OCR (Tesseract), SQLite, Power BI, Tableau
Cloud & deployment: AWS (Cloud Foundations), Docker, Docker Compose, FastAPI, Django, Flask, Spring Boot, Symfony, REST APIs, n8n, Git
Methods: Microsoft TDSP, CRISP-DM, Design Thinking, Agile/Scrum

CERTIFICATIONS

- AWS Cloud Foundations — Amazon Web Services (badge, Nov 2026)
- Fundamentals of Deep Learning — NVIDIA (2026)
- Applications of AI for Anomaly Detection — NVIDIA (2026)
- Building Transformer-Based Natural Language Processing Applications — NVIDIA (2026)
- CCNA 2: Switching, Routing, and Wireless Essentials (SRWE) — Cisco Networking Academy (2025)

LANGUAGES

English: fluent | French: advanced (professional working proficiency) | Arabic: native